WIRTINGER'S INEQUALITY

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Introduction

I only recently learned that Wirtinger's Inequality, which provides an upper bound on the square integral of a C^1 function f (of average value zero) over $[0,T]$, can be used to prove the Isoperimetric Inequality. This was shown by Hurwitz in 1901: [Hur01]. This article proves the Wirtinger inequality using Parseval's Theorem, but readers should note that there are multiple ways to prove the result. I do not go into the proof of the Isoperimetric Inequality to keep the article brief; one may invoke Green's Theorem to obtain a clean proof in $\mathbb{R}^2$: [Wik26].

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Proof

Wirtinger's Inequality.

Let f be C^1 over the interval $[0,T]$ with $\int_0^T f(t)\,dt = 0$, and $f(0) = f(T)$, then we have the estimate

$$\int_0^T |f(t)|^2\,dt \leq \frac{T^2}{4\pi^2}\int_0^T |f'(t)|^2\,dt.$$

Proof: We note that $\hat{f}(0) = \frac{1}{T}\int_0^T f(t)\,dt = 0$. Moreover, the Fundamental Theorem of Calculus implies

$$\hat{f}'(0) = \frac{1}{T}\int_0^T f'(t)\,dt = \frac{1}{T}(f(T) - f(0)) = 0.$$

Suppose $n \neq 0$. Then the Fourier coefficients of f are related to those of f' by the following calculation:

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$$\begin{aligned}
\hat{f}(n) &= \frac{1}{T}\int_0^T f(t)e^{-\frac{2\pi i n t}{T}}\,dt \\
&= \frac{1}{T}\left(\frac{T}{2\pi i n}\left[f(t)e^{-\frac{2\pi i n t}{T}}\right]_0^T + \frac{T}{2\pi i n}\int_0^T f'(t))e^{-\frac{2\pi i n t}{T}}\,dt\right) \\
&= \frac{1}{2\pi i n}\int_0^T f'(t))e^{-\frac{2\pi i n t}{T}}\,dt \\
&= \frac{T}{2\pi i n}\left(\frac{1}{T}\int_0^T f'(t)e^{-\frac{2\pi i n t}{T}}\,dt\right) \\
&= \frac{T}{2\pi i n}\hat{f}'(n).
\end{aligned}$$

Therefore, for $n \neq 0$ we have

$$|\hat{f}(n)|^2 = \frac{T^2}{4\pi^2 n^2}|\hat{f}'(n)|^2 \leq \frac{T^2}{4\pi^2}|f'(n)|^2.$$

Invoking Parseval's Identity (recall that $\hat{f}(0) = \hat{f}'(0) = 0$), we obtain

$$\begin{aligned}
\frac{1}{T}\int_0^T |f(t)|^2\,dt = \sum_{n=-\infty}^{\infty}|\hat{f}(n)|^2 = \sum_{n\neq 0}|\hat{f}(n)|^2 \leq \frac{T^2}{4\pi^2}\sum_{n\neq 0}|\hat{f}'(n)|^2 \\
\leq \frac{T^2}{4\pi^2}\left(\frac{1}{T}\int_0^T |f'(t)|^2\,dt\right),
\end{aligned}$$

so we obtain the final result that

$$\int_0^T |f(t)|^2\,dt \leq \frac{T^2}{4\pi^2}\int_0^T |f'(t)|^2\,dt,$$

completing the proof. ■

References

[Hur01] Adolf Hurwitz. "Sur le problème des isopérimètres". In: *Comptes Rendus des Séances de l'Académie des Sciences* 132 (1901). JFM 32.0386.01, pp. 401–403. URL: `https://gallica.bnf.fr/ark:/12148/bpt6k30888.f429`.

[Wik26] Wikipedia contributors. *Wirtinger's inequality for functions*. 2026. URL: `https://en.wikipedia.org/wiki/Wirtinger's_inequality_for_functions` (visited on 08/29/2026).